

Seat Belt Violation Detection & License Plate Logging System

Objective

The objective of this project is to design and implement an intelligent traffic monitoring system that detects drivers, classifies them into Person_with_seat_belt and Person_without_seatbelt, identifies violations, detects license plates, extracts plate text using OCR, stores data in a database, and provides a user interface.

Workflow Pipeline

Input Image/Video → YOLO Model Detection (Person_with_seatbelt, Person_without_seatbelt)

- Identify all persons
- Filter persons without seat belt
- For each violating person:
 - YOLO License plate detection model
 - Find nearest license plate with the violating person
 - Crop plate region
 - Apply OCR
 - Extract text
- Store results in database
- Display output in UI

Dataset Links

Seat Belt Detection Dataset: <https://universe.roboflow.com/reduka-ed2ia/seat-belt-detection-c2nia>

License Plate Dataset: <https://universe.roboflow.com/manasa-n-n/car-license-plate-detection-maehj>

Project Steps

Dataset Preparation: Download and organize datasets in YOLO format.

Train YOLO Model for detecting Person_with_seatbelt, Person_without_seatbelt,

Train another Model for License_plate Detection.

Run inference on images/videos.

Identify persons without seatbelt.

If person without seatbelt than run YOLO License Plate Detection model on image

Find nearest license plate to the Person without seatbelt using bounding box matching.

Crop license plate region.

Apply OCR to extract plate number.

Store results in database (use PostgreSQL).

Build UI using Gradio.

Workflow Pipeline:

Input Image / Video



YOLO Model 1 (Detection)

├── Person_with_seatbelt

├── Person_without_seatbelt



Step 1: Identify all detected persons



Step 2: Filter persons WITHOUT seatbelt



Step 3: For each violating person:

├── YOLO License plate detection model (YOLO Model 2)

├── Find nearest license plate to the person with no seatbelt

├── Crop plate region

├── Apply OCR (Use Tesseract OCR)

└── Extract text



Step 4: Store results in database



Step 5: Display output in UI (Gradio)



Step 6: deploy on HuggingFace

Database Integration

Field	Description
plate_number	Extracted text
violation_type	No Helmet